

Fortgeschrittenenpraktikum

Raman Spectroscopy

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Abstract

The vibration Raman effect consists of the appearance of vibration inelastic (Stokes) and superelastic (Antistokes) lines in the fluorescence spectrum of gases, liquids and solids. In this experiment the vibration Raman spectra of various liquid substances are recorded and identified.

1 Theoretical background

1.1 Raman scattering

Vibrational spectroscopies are employed in the determination of the molecular structures and are used in many areas. Among the available techniques, Raman spectroscopy is essential in the study of nanostructured materials and biological systems. Detailed information of the molecular structure can be obtained, because each vibration of atoms shows a characteristic position and intensity. Both position and intensity are influenced by the chemical bonds, inter- and intramolecular forces [1]

An inelastic scattering process is responsible for the appearance of vibrational bands (Raman spectra). Unlike in IR spectroscopy (where the photon energy must coincide with the energy difference between two available vibrational states), in Raman spectroscopy a molecule can scatter a monochromatic radiation. The Raman spectrum appears in a wavelength λ slightly higher or lower than the incident radiation [1].

Thus Raman scattering can be visualized as a perturbation of the molecule by incident photon, which eigenstates can be described as a linear combination of the stationary states (or time-independent) wave functions of the molecule. The photon released when the molecule returns to electronic ground state has, more often, the same incident photon energy (Rayleigh scattering), but Raman scattering (which contains the vibrational information) can also be present. [1]

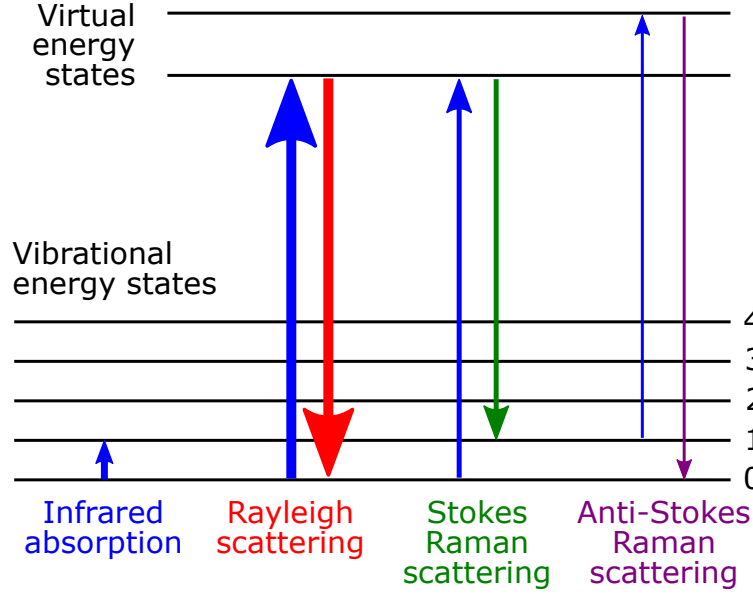


Figure 1: Energy-level diagram showing the states involved in Raman spectra [2]

A vibration (vibration coordinate ρ) is *Raman active* when the polarizability α (or *induced dipolar momentum*) of the molecule is changed (described as $\frac{\partial \alpha}{\partial \rho} \neq \emptyset$). This illustrates that the nature of the Raman effect is physically different from the infrared. Selection rules are also different and thus lead to a different spectra [1].

1.2 Lock-in amplifiers

A lock-in amplifier is a type of amplifier that extracts a signal with a known carrier wave from a noisy environment. Depending on the instrument, signals up to orders of magnitude smaller than the noise components, potentially close by in frequency, can still be reliably detected [3]. This reliability depends on the *dynamic reserve* of the instrument, which is simply a measure of how well the instrument can extract the signal when there's a much larger unwanted noise sitting on top of it. Modern instruments achieve a dynamic reserve up to 120 dB [3].

For amplitude quantities,

$$\text{dB} = 20 \log_{10} \left(\frac{A_{\text{noise}}}{A_{\text{signal}}} \right) \quad (1)$$

So for 120 dB, the above equation yields

$$120 \text{ dB} = 20 \log_{10} \left(\frac{A_{\text{noise}}}{A_{\text{signal}}} \right) \quad (2)$$

which gives

$$\frac{A_{\text{noise}}}{A_{\text{signal}}} = 10^{\frac{120}{20}} = 10^6 \quad (3)$$

So modern instruments can theoretically recover signals with an order of magnitude 10^{-6} smaller than the noise.

A lock-in amplifier performs a multiplication of its input with a reference signal, also sometimes called down-mixing or heterodyne/homodyne detection, and then applies an adjustable low-pass filter to the result [4]. This method is termed demodulation or phase-sensitive detection and isolates the signal at the frequency of interest from all other frequency components. The reference signal is either generated by the lock-in amplifier itself or provided to the lock-in amplifier and

the experiment by an external source. The reference signal is usually a sine wave but could have other forms, too. Demodulation with a pure sine wave enables selective measurement at the fundamental frequency or any of its harmonics [4].

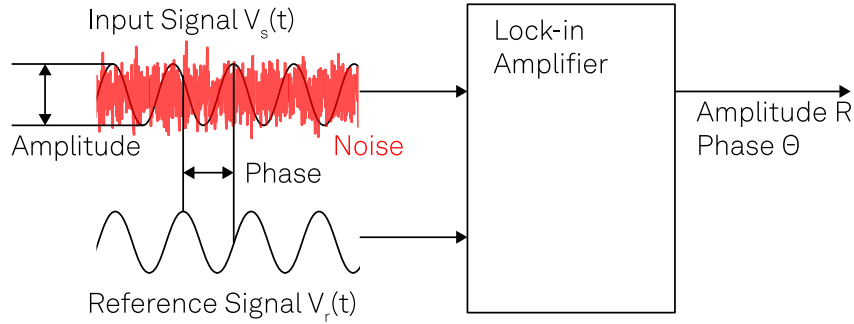


Figure 2: Lock-in amplifiers are capable of measuring the amplitude and the phase of a signal relative to a defined reference signal, even if the signal is entirely buried in noise. Adapted from [4]

1.3 Classical and Quantum Mechanical Theory

The classical picture of Raman scattering starts with the electric field of the incident laser light,

$$E(t) = E_0 \cos(\omega_0 t), \quad (4)$$

which induces a dipole moment in the molecule,

$$p(t) = \alpha E(t), \quad (5)$$

where α is the molecular polarisability.

Since the molecule is vibrating, its polarisability can change with the normal coordinate Q of the vibration:

$$\alpha(Q) = \alpha_0 + \left(\frac{\partial \alpha}{\partial Q} \right)_0 Q. \quad (6)$$

If the vibration has frequency ω_v , the induced dipole contains components at

$$\omega_0, \quad \omega_0 - \omega_v, \quad \omega_0 + \omega_v. \quad (7)$$

These correspond to:

- **Rayleigh scattering:** ω_0
- **Stokes Raman scattering:** $\omega_0 - \omega_v$
- **anti-Stokes Raman scattering:** $\omega_0 + \omega_v$

In the quantum mechanical picture, the incident photon interacts with the molecule through a virtual state. The molecule changes its vibrational energy, and therefore the scattered photon has a different energy:

$$E_{\text{out}} = E_{\text{in}} \mp \hbar \omega_v. \quad (8)$$

Thus, the Raman shift directly provides information about the vibrational energy levels of the molecule.

1.4 Polarisability

Polarisability describes how easily the electron cloud of a molecule can be distorted by an external electric field. The induced dipole moment is given by

$$\mathbf{p}_{\text{ind}} = \boldsymbol{\alpha}\mathbf{E}, \quad (9)$$

where $\boldsymbol{\alpha}$ is generally a tensor.

The important Raman selection rule is

$$\left(\frac{\partial \alpha}{\partial Q} \right)_0 \neq 0 \quad (10)$$

A vibrational mode is therefore **Raman active if the vibration produces a change in the molecular polarisability**.

1.5 Degree of Depolarisation

The degree of depolarisation describes how much the polarisation of the scattered Raman light differs from that of the incident laser light. It is commonly defined as

$$\rho = \frac{I_{\perp}}{I_{\parallel}}, \quad (11)$$

where $I_{\perp}$ is the intensity of the Raman light polarised perpendicular to the incident polarisation and $I_{\parallel}$ is the intensity parallel to it.

For the usual treatment of Raman scattering from randomly oriented molecules,

$$0 \leq \rho \leq \frac{3}{4}. \quad (12)$$

A low value of ρ corresponds to strongly polarised Raman scattering, whereas a larger value indicates stronger depolarisation. The degree of depolarisation can provide information about the symmetry of a vibrational mode.

1.6 Scattering Cross Section

The scattering cross section σ describes the probability that an incident photon will be scattered by a molecule. It can be interpreted as an effective area that characterises the scattering probability.

Raman scattering has a very small scattering cross section compared with Rayleigh scattering. Consequently, Raman signals are weak and require sensitive detectors and sufficiently strong laser illumination.

1.7 Normal Coordinates and Normal Oscillations

A molecule can undergo many different vibrational motions. These motions can be decomposed into independent **normal modes**.

Each normal mode has:

- a characteristic pattern of atomic motion,
- a characteristic vibrational frequency, and
- a normal coordinate Q describing the amplitude of the vibration.

The molecular motion can therefore be described using a set of normal coordinates,

$$Q_1, Q_2, Q_3, \dots \quad (13)$$

which correspond to the different normal modes of vibration.

The dependence of the polarisability on a normal coordinate is particularly important for Raman spectroscopy:

$$\alpha(Q) = \alpha_0 + \left(\frac{\partial \alpha}{\partial Q} \right)_0 Q + \dots \quad (14)$$

1.8 Induced and Permanent Dipole Moment

A molecule may possess a **permanent dipole moment** due to an asymmetric distribution of electric charge,

$$\mathbf{p}_0 \neq 0. \quad (15)$$

An external electric field can also distort the electron cloud and produce an **induced dipole moment**:

$$\mathbf{p}_{\text{ind}} = \alpha \mathbf{E}. \quad (16)$$

Raman scattering originates from this induced dipole. The electric field of the laser drives the induced dipole, which subsequently radiates the scattered light.

1.9 Raman shift wavenumber

Given that energy shifts in the incident photons are not wavelength dependent, it makes sense to express the raman spectra in a wavelength-independent notation. Raman shift wavenumbers are the unit used to do so. The raman shift of a photon is simply

$$\Delta\nu = \nu_0 - \nu_s \quad (17)$$

where ν_0 is the frequency of the incident light and ν_s is the frequency of the scattered light.

This way we can express raman shifts without involving laser wavelengths knowing well that raman spectra will show in the same wavenumber regardless of the laser used.

2 Methodology

2.1 Calibration of the Monochromator

First the monochromator's transmission curves for horizontally and vertically polarized light were looked after. To do so, a Quartz Tungsten-Halogen Lamp was used. This lamp features a stable output and broadband, incoherent illumination in the visible and near-IR portions of the spectrum.

A polarizer was placed at the input of the monochromator as show in figure 3. For further analysis, it is important that both transmission curves are comparable, and hence measured with the same set parameters in the experiment (or somehow made compatible through data processing). After tweaking the parameters of the coupler and the photomultiplier high voltage supply, the following configuration was found to have enough dynamic range to record both horizontal and vertical without saturating the instrument signals: The setup was first manipulated as to fit the output voltage obtained by horizontally polarized light best, just below the saturation point for the whole spectra. The coupler was set in 10^{-6} , the voltage supply for the photomultiplier to ~ 300 V and the spectra was taken from 400 nm to 800 nm at a speed of 5 nm/s.

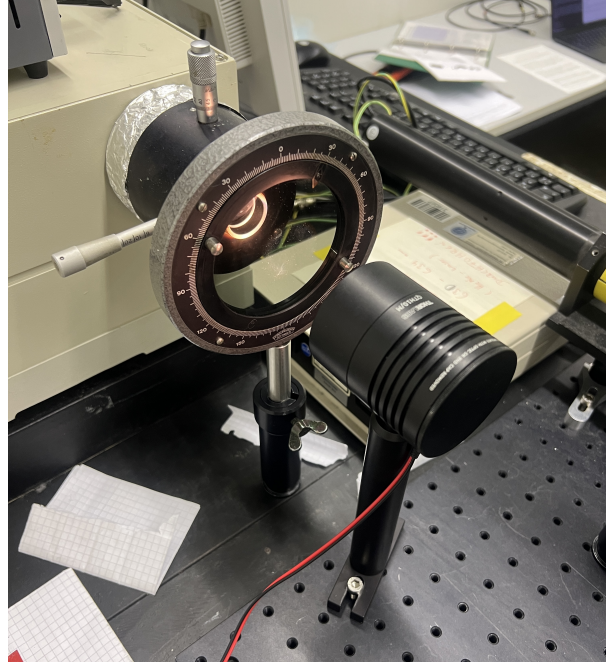


Figure 3: Halogen lamp, polarizer and monochromator input

Then, with the same set of parameters, the polarizer was vertically aligned and once again the spectra was taken from 400 nm to 800 nm at the same speed.

The curve corresponding to horizontal polarization can be seen in figure 4. The curve corresponding to vertical polarization can be seen in figure 5. Both of these figures are raw images extracted from the TDS2001IC oscilloscope screen.

In the experiment, the monochromator grating is adjusted by a servo motor connected to a microcontroller which was programmed to make precise adjustments to the grating. In order to know which voltage corresponds to which wavelength, the arduino's voltage output is also recorder in the oscilloscope to match the data.

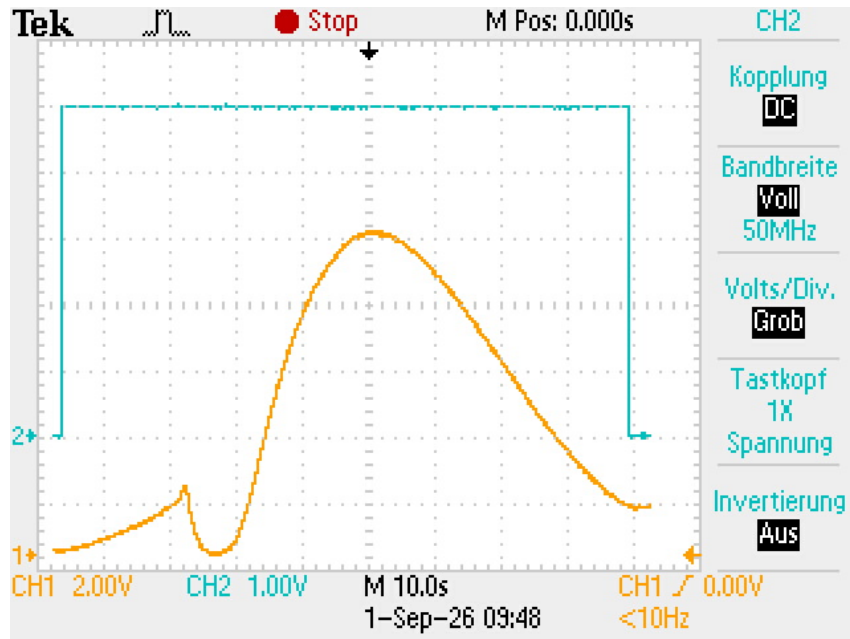


Figure 4: Voltage curve of the photomultiplier for horizontally polarized light (orange) and servo voltage (blue).

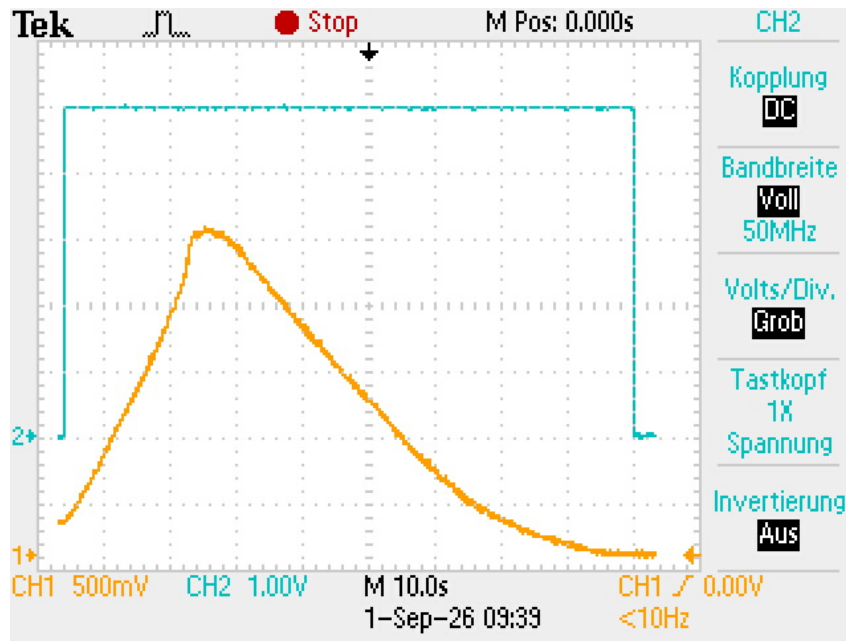


Figure 5: Voltage curve of the photomultiplier for vertically polarized light (orange) and servo voltage (blue).

The data shown in both figures 4 and 5 will be useful to apply instrument response correction (correct the actual raman spectra taking into account the wavelength-dependent intensity of the photocromator).

The Halogen lamp was removed from the array, and the two lenses that form the telescope were placed so that the laser light goes in as focused as possible in the monochromator. The ethanol + salt sample was inserter in the chamber. This sample has a very strong rayleigh reflection, and is thus the ideal sample to align the experiment. The monochromator was swept in between 622 and 642 nm. The Voltage curve can be seen in figure 6. Note that the lock-in mechanism and the chopper are not active yet.

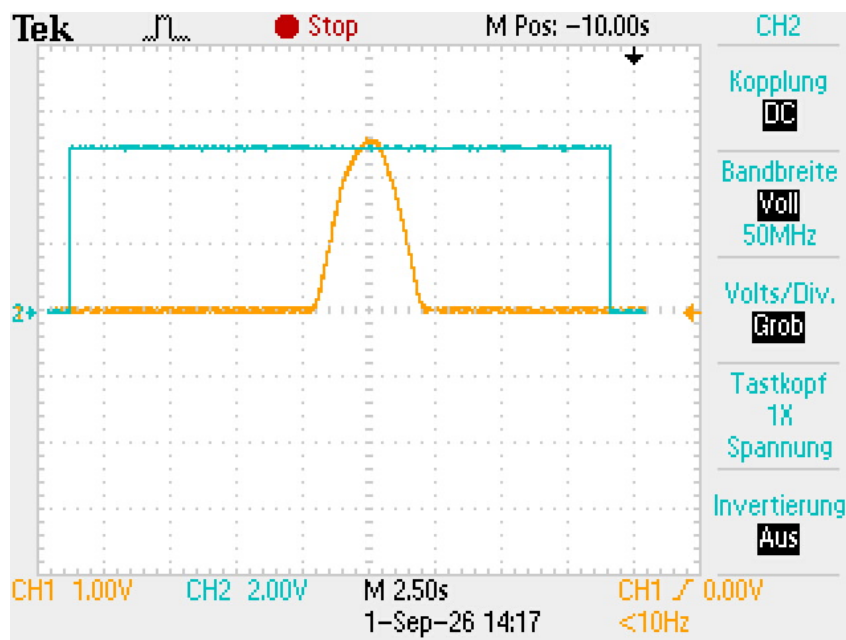


Figure 6: Voltage curve of the 633 nm laser (ethanol +salt cuvette)

Figure 6 shows that our laser indeed has a gaussian profile with a peak presumably at 633 nm.

2.1.1 instrument response correction

Figures 4 and 5 prove that the monochromator has a wavelength-dependent response $R(\lambda)$, thus we can say that approximately

$$I_{measured}(\lambda) = I_{true}(\lambda)R(\lambda) \quad (18)$$

So to correct the measured spectrum we divide by the response of the monochromator.

2.2 Experiment Setup

Having the monochromator response, the rest of the setup can be properly introduced in the experiment. A laser of wavelength of 633 nm goes through a mechanical chopper whose frequency can be adjusted by a control box. The laser then goes through a cage system containing a sample, it is then reflected onto the telescope system. Finally the laser goes inside the photochromator. the photomultiplier then outputs a signal for data analysis. In between the sample and the monochromator, a polarizer is introduced to analyze the change in light polarization after the sample. The laser polarization can also be adjusted by rotating the laser tube.

A diagram of the experiment can be seen in 7.

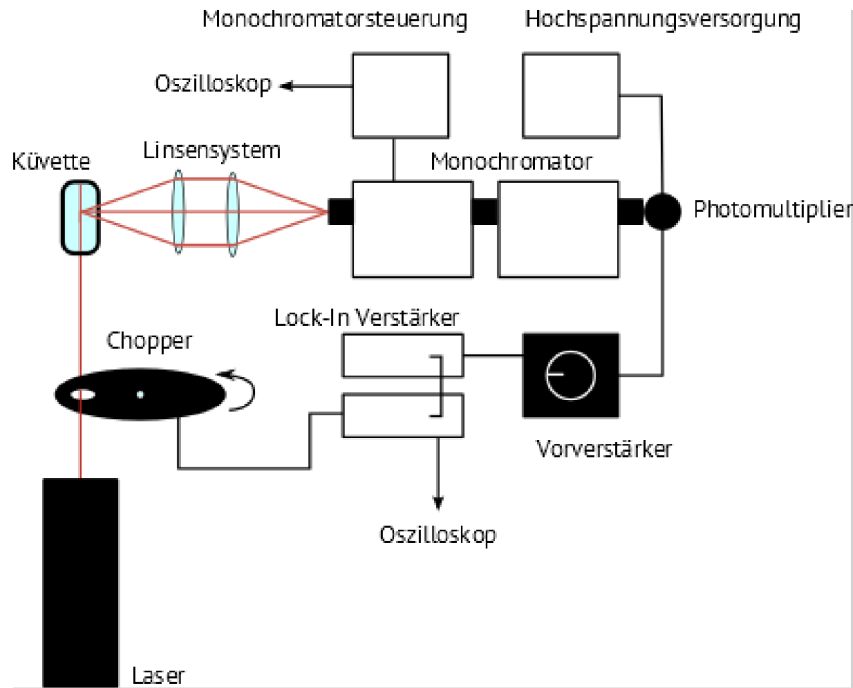


Figure 7: Setup of the experiment. Taken from [5]

After the monochromator, the signal processing electronics take place. Given that the photomultiplier signal is very small, immediately after a preamp is connected to strengthen the signal. Said signal is then fed to a phase sensitive Detector. This detector has as reference the phase-shifted signal of the chopper. A picture of this part of the array is seen in figure 8. These electronics and configuration is explained in detail in section 2.3.1

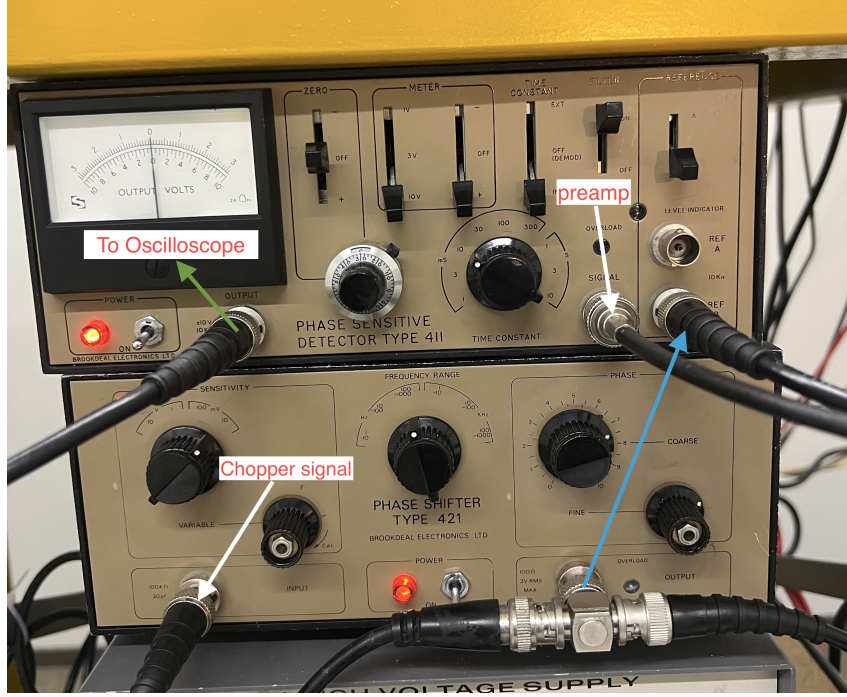


Figure 8: Lock-In amplifier inputs and outputs (knobs are not in final position)

2.3 Lock-In Amplifier

It will be seen in the following sections of this protocol that raman peaks are of very low intensity and can be hard to discern in a regular spectrum; this is because raman scattering is considerably weaker than rayleigh scattering. Often they can be buried in noise and the rayleigh peaks can also bury them if they're overlapping.

To overcome this challenge, we need a way to remove unwanted noise from the spectra. This is where the Lock-In amplifier comes in handy as explained in section 1.2

2.3.1 Lock-in amplifier configuration

The optical chopper was set to an arbitrary frequency (around 130 Hz). The chopper driver generates an AC output that was forwarded to our Phase shifter. The output of the phase shifter is then taken as a reference signal in the phase sensitive detector (PSD). Furthermore, the chopper frequency can be calculated with the AC output signal. The output of the current amplifier (whose input is the photomultiplier) is then taken as the input signal on the PSD. Refer to figure 8 for a visual image of the connections.

The fine-tuning of the lock-in mechanism took place. Given that the chopper has a frequency of around 130 Hz, the phase shifter frequency has to be set accordingly to the range 100 – 1000 Hz. Then the phase shift is adjusted either to match closely either 0° or 180°. This is because the photodiode signal at the chopping frequency has the form

$$S(t) = A \cos(\omega t + \phi) \quad (19)$$

while the reference from the chopper is

$$R(t) = \cos(\omega t + \theta) \quad (20)$$

At the PSD, both signals are multiplied and low-pass filtered, giving an expression resembling

$$V(t) = \frac{A}{2} \cos(\phi - \theta) \quad (21)$$

It is evident that the resulting signal has an absolute maxima when the above condition is met. In practice, this is done by hooking the photodiode and the phase-shifted chopper in the two channels of the oscilloscope and adjusting the time scale to distinguish the square oscillating signal as seen in figure 9

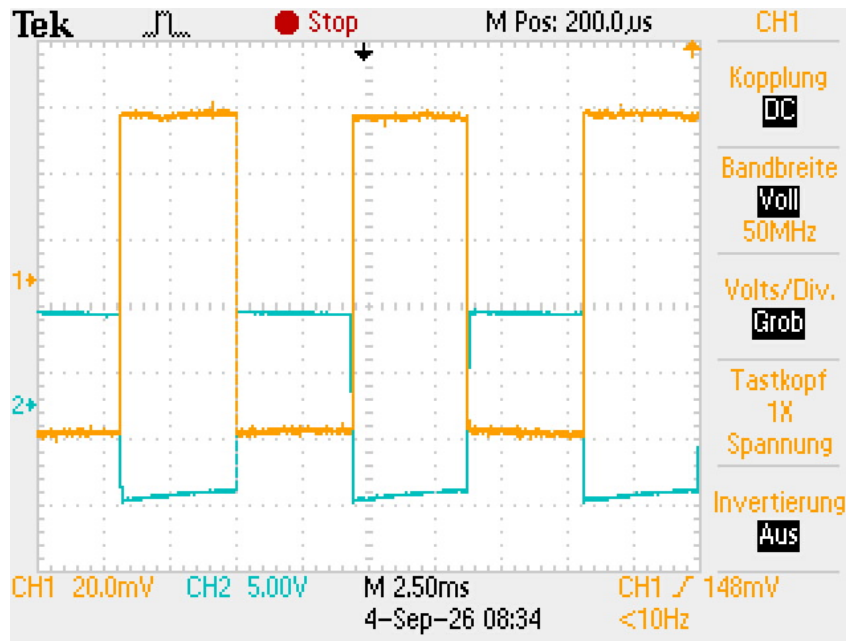


Figure 9: Phase shifted chopper (blue) and photodiode (yellow) on the oscilloscope. They fulfill the locked condition

The time constant α of the PSD was introduced after that. The time constant is the low pass constant which determines how quickly the PSD output is allowed to respond to changes by controlling the bandwidth of its low pass filter.

Simply put, if α is small, the output responds quickly, but more noise gets through. Viceversa, if α is big, output changes slowly, but more noise is rejected.

3 Results

After processing the data by taking into account the servo response, both of the curves seen in figures 4 and 5 are now shown with the wavelength (nm) in figures 10 and 11

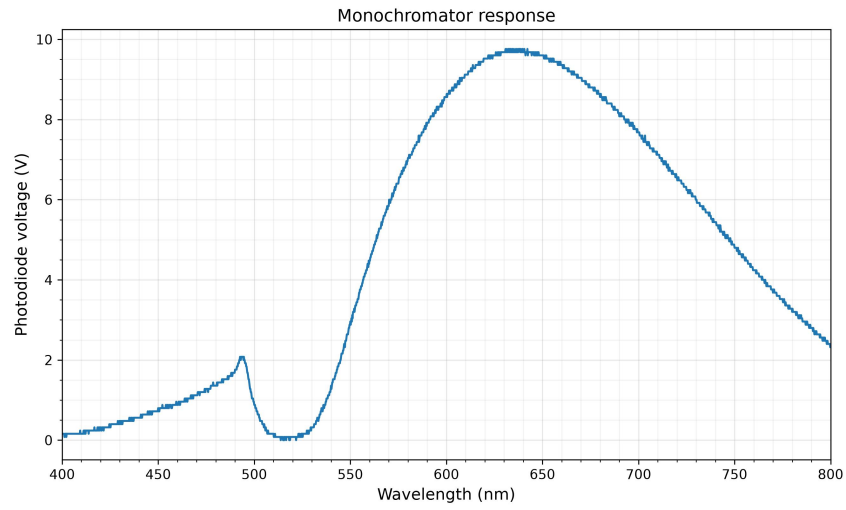


Figure 10: Monochromator response (horizontally polarized light).

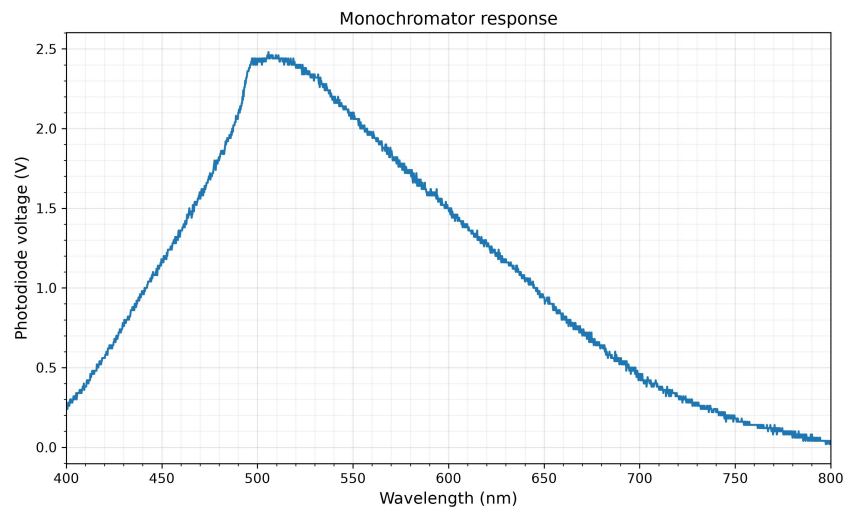


Figure 11: Monochromator response (vertically polarized light).

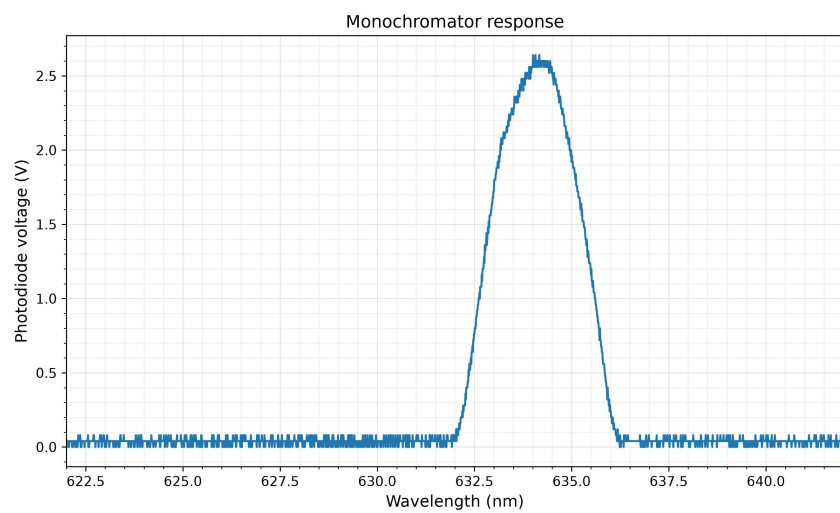


Figure 12: Laser spectra (ethanol + salt)

Take note that all of the next results are corrected to consider the monochromator polarizability/wavelength response as explained in section 2.1.1.

3.1 Raman spectra of CCL4 and Benzene

As suggested in section 1.9, all of the results are presented in the appropriate raman shift wavenumber.

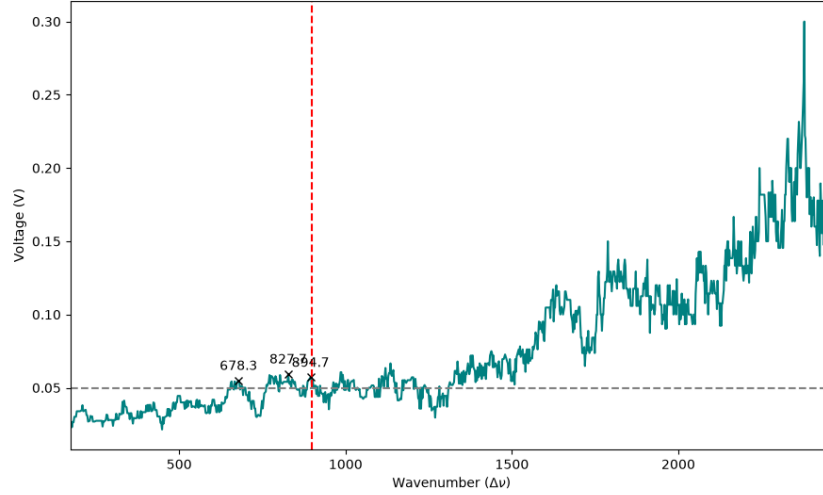


Figure 13: CCL4 spectra with laser at 0° and polarizer at 0°

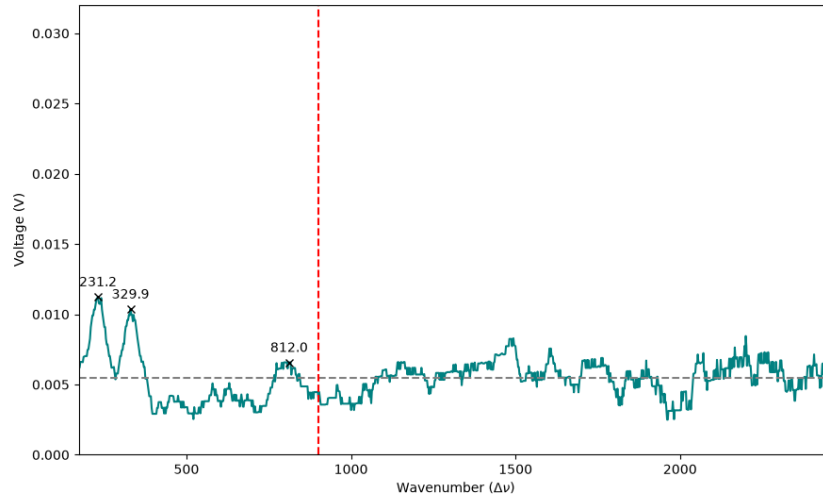


Figure 14: CCL4 spectra with laser at 0° and polarizer at 90°

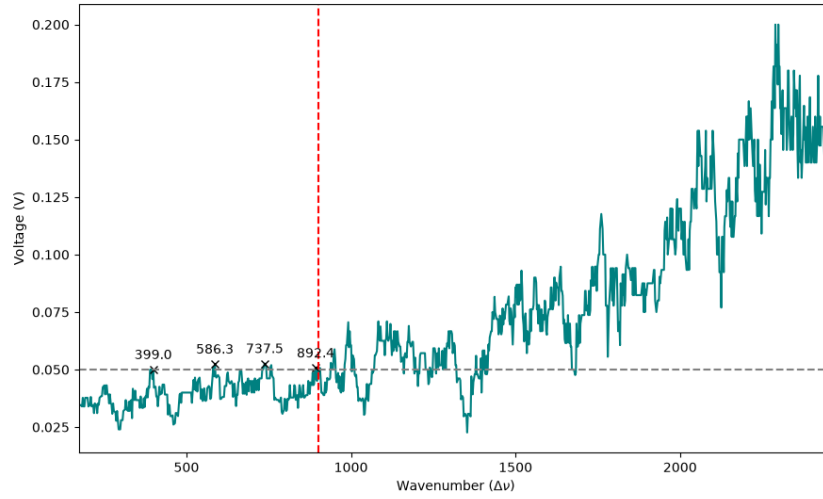


Figure 15: CCL4 spectra with laser at 90° and polarizer at 0°

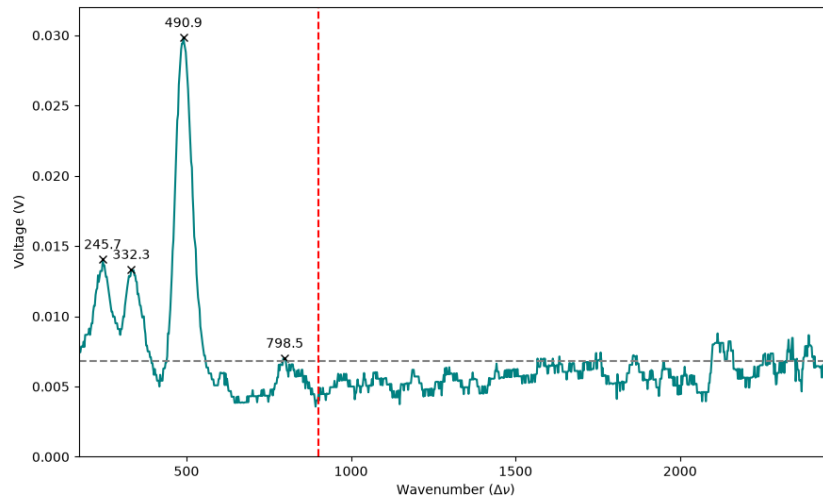


Figure 16: CCL4 spectra with laser at 90° and polarizer at 90°

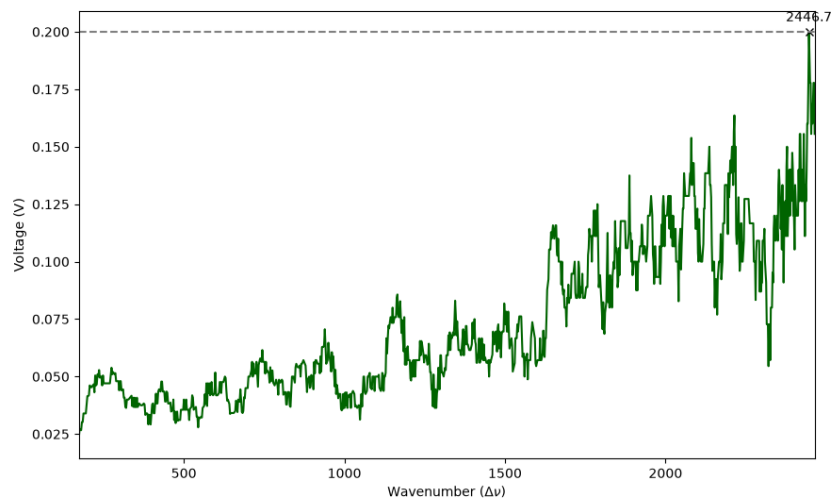


Figure 17: Benzene spectra with laser at 0° and polarizer at 0°

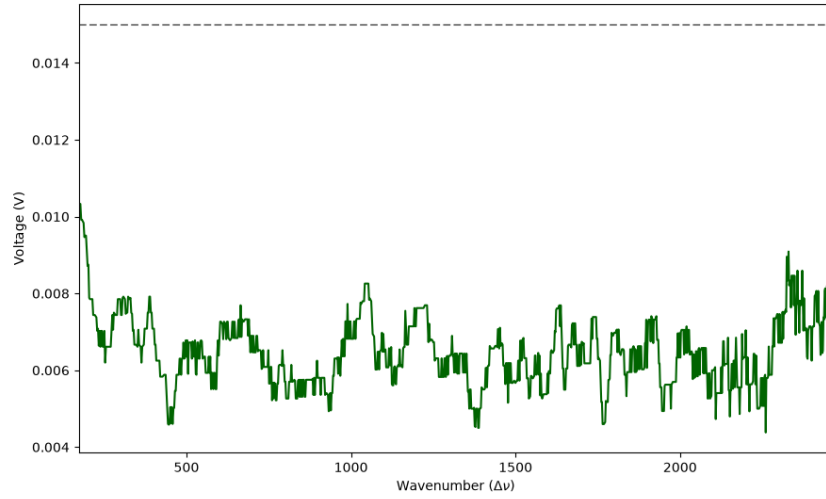


Figure 18: Benzene spectra with laser at 0° and polarizer at 90°

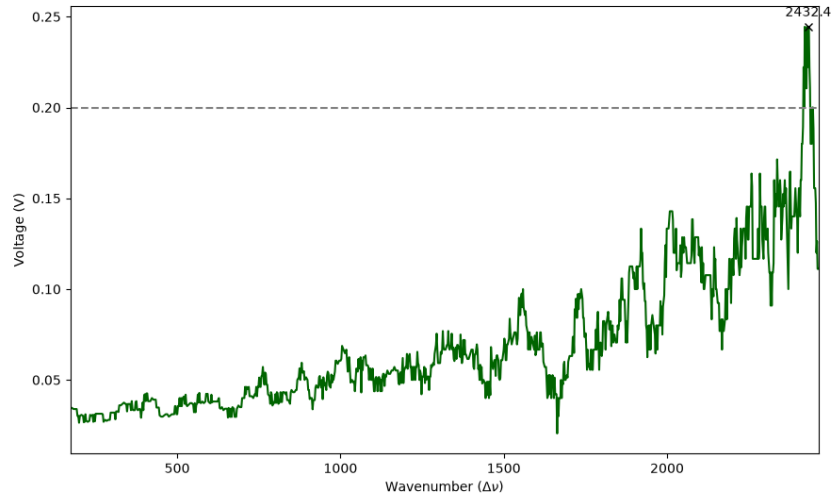


Figure 19: Benzene spectra with laser at 90° and polarizer at 0°

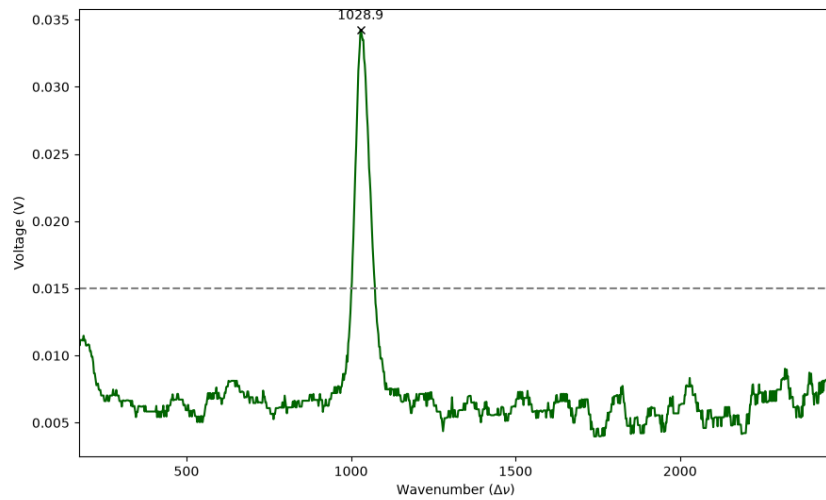


Figure 20: Benzene spectra with laser at 90° and polarizer at 90°

3.2 Unknown sample Spectra

Figures 21 and 22 show the spectra recorded for two samples whose content was not known in advance.

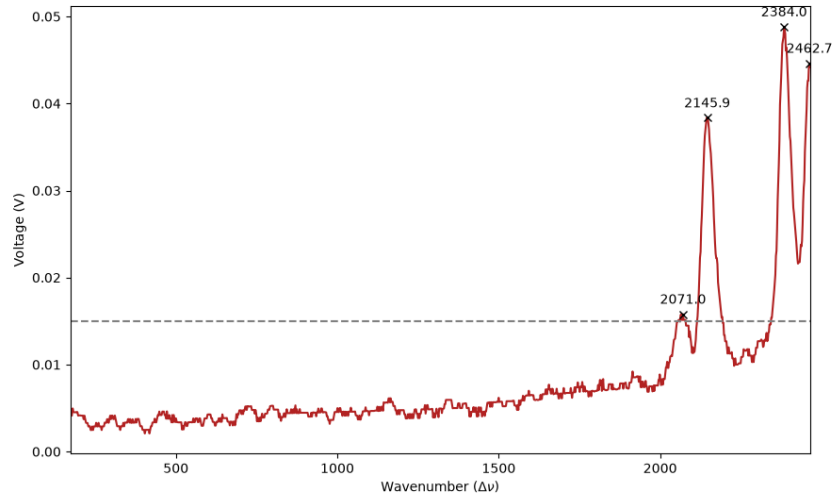


Figure 21: Spectra of *unknown sample 1*

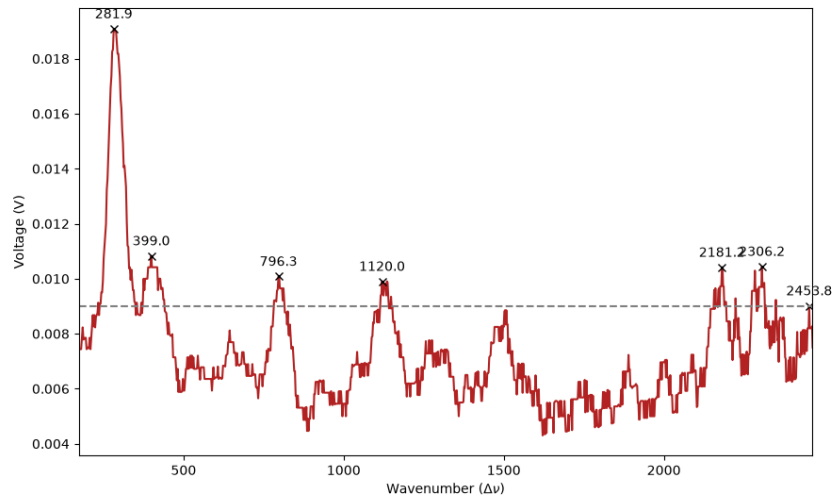


Figure 22: Spectra of *unknown sample 2*

3.3 Ethanol concentration of an unknown sample

Spectra of multiple samples with known ratios of ethanol and distilled water were recorded during the experiment and are shown in figures 23, 24, 25, and 26.

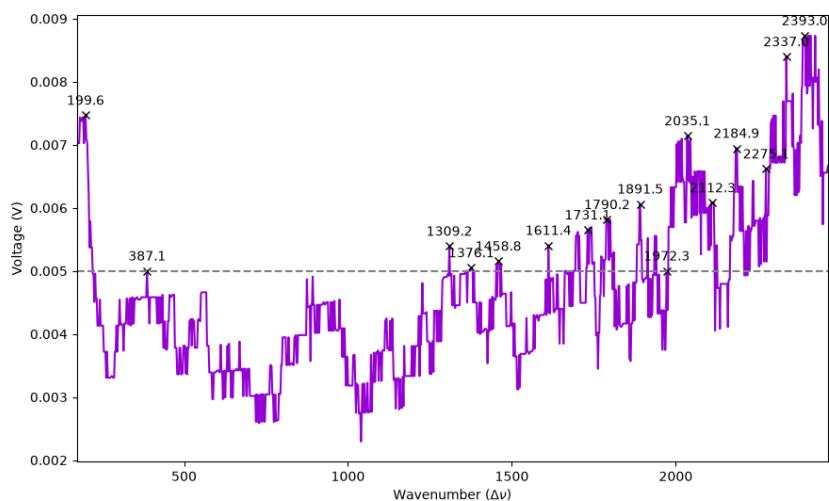


Figure 23: Spectra of distilled water.

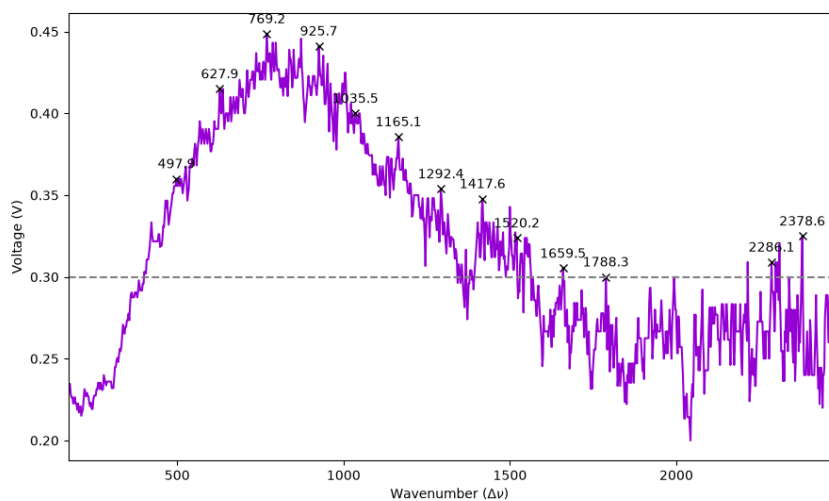


Figure 24: Spectra of ethanol at 50%

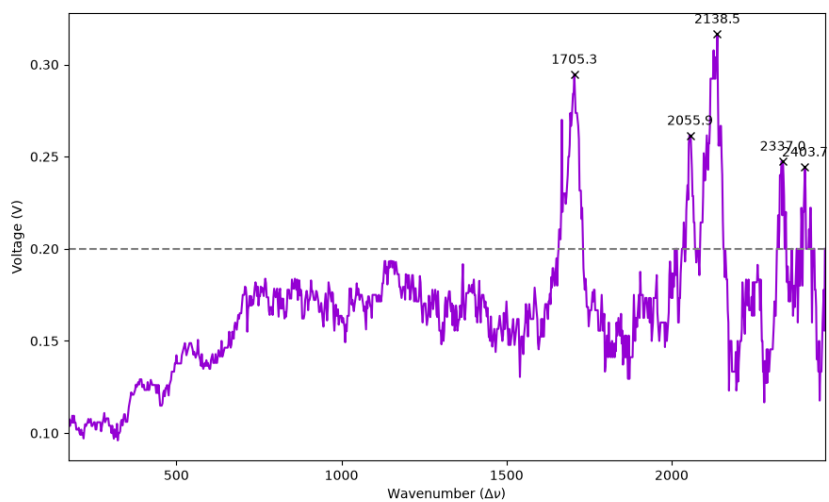


Figure 25: Spectra of ethanol at 75%

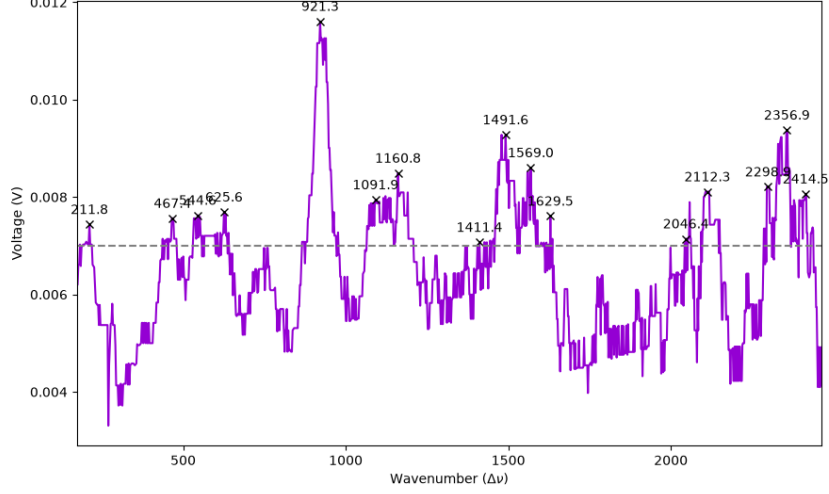


Figure 26: Spectra of ethanol at 100%

3.4 Fit of relevant peaks

3.4.1 CCL4

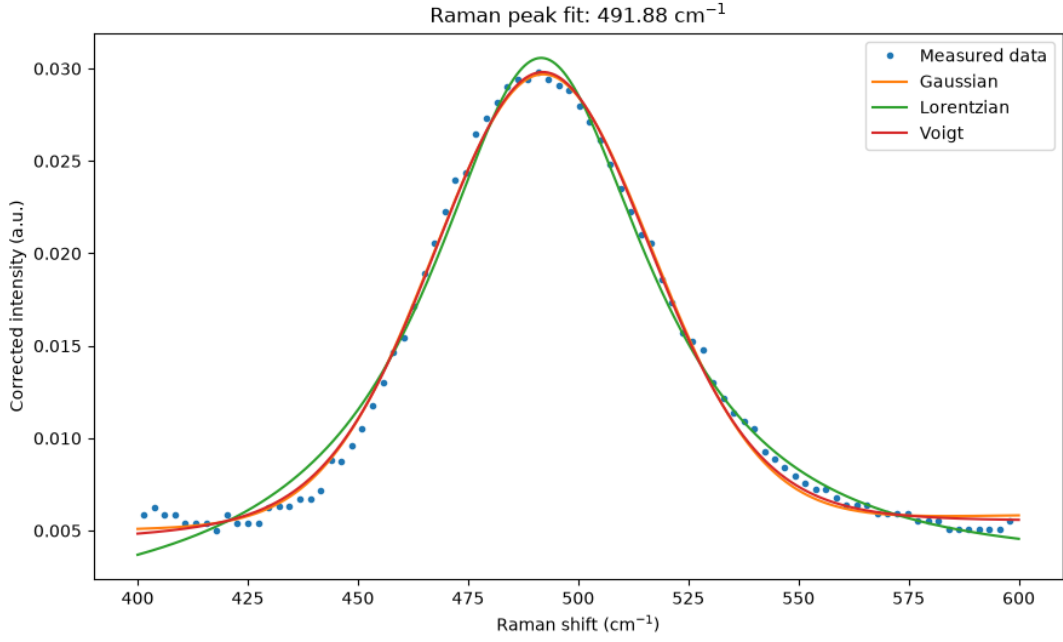


Figure 27: Fit for Raman peak at 491.9 cm^{-1}

Peak	Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	$\lambda \text{ (nm)}$	FWHM (cm^{-1})	R^2
491	Gaussian	492.0 ± 0.2	653.35 ± 0.01	58.2	0.9954
491	Lorentzian	491.4 ± 0.3	653.32 ± 0.01	61.8	0.9902
491	Voigt	491.9 ± 0.2	653.34 ± 0.01	58.8	0.9956

Table 1: Fit parameters for the 491.9 cm^{-1} Raman peak.

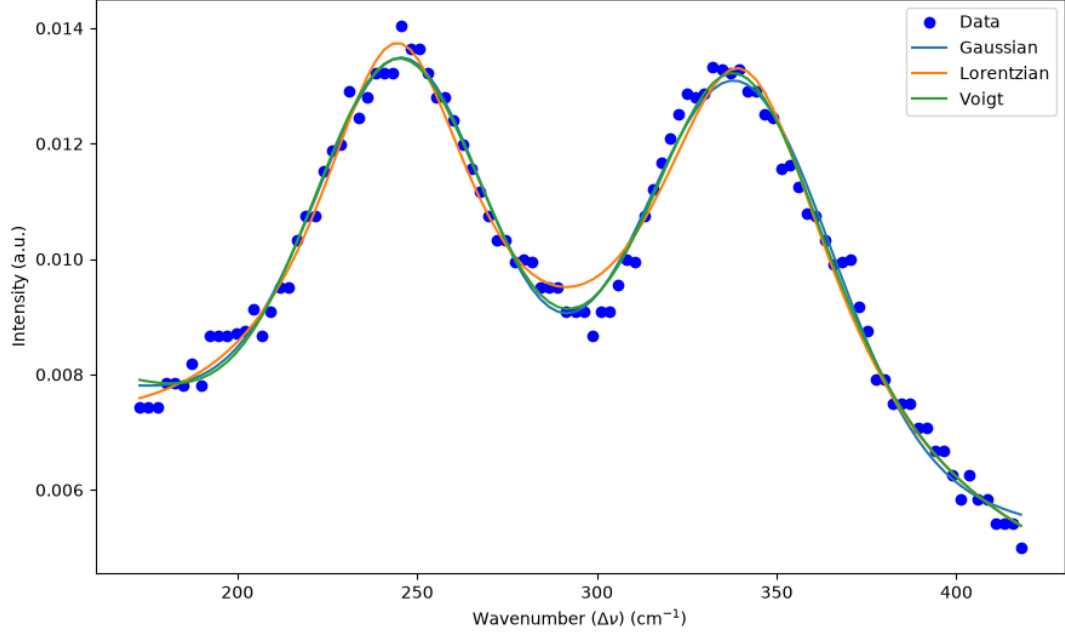


Figure 28: Fit for Raman peak at 491.9 cm^{-1}

Peak	Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	FWHM (cm ⁻¹)	R^2
245	Gaussian	246.2 ± 0.5	56.0	0.9838
332	Gaussian	339.0 ± 0.4	62.0	0.9838
245	Lorentzian	244.2 ± 0.5	56.1	0.9806
332	Lorentzian	340.5 ± 0.5	64.4	0.9806
245	Voigt	245.0 ± 0.6	52.4	0.9848
332	Voigt	338.7 ± 0.8	67.5	0.9848

Table 2: Fit parameters for the twin peaks 245 cm^{-1} and 332 cm^{-1} Raman peak.

3.4.2 Benzene

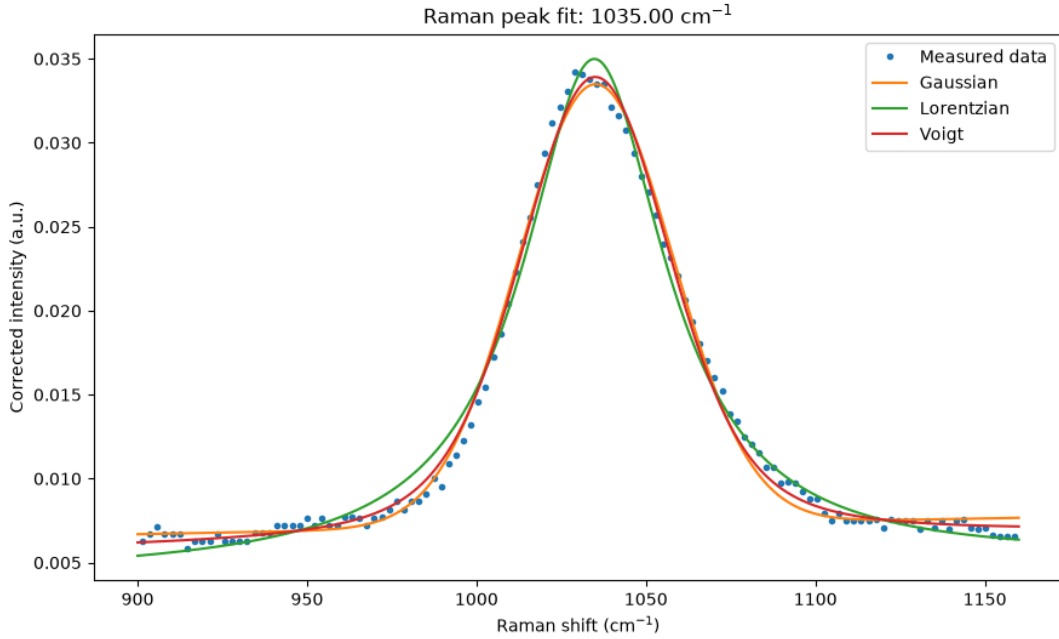


Figure 29: Fit for Raman peak at 1035 cm^{-1}

Peak	Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	$\lambda \text{ (nm)}$	FWHM (cm^{-1})	R^2
1028	Gaussian	1035.1 ± 0.2	677.38 ± 0.01	53.8	0.9934
1028	Lorentzian	1034.7 ± 0.3	677.37 ± 0.01	51.4	0.9892
1028	Voigt	1035.0 ± 0.2	677.38 ± 0.01	53.2	0.9950

Table 3: Fit parameters for the Benzene 1035 cm^{-1} Raman peak.

4 Discussion

4.1 CCl4

4.2 Benzene

4.3 CCL4 and Benzene

4.3.1 peaks

Tables 4 and 5 show the reported raman shifts according to [6] against the measured shifts in the experiment.

For CCL4, we were able to record all but one of the modes reported in [6]. This is likely because the denegerate stretch located at 761.7 cm^{-1} is close enough to 790.4 cm^{-1} , and our alignment couldn't provide enough output in the photomultiplier channel to resolve efficiently this two peaks, but are instead hidden in one another's intensity. Another likely explanation was the sweep speed in our measurements. We opted for a sweep speed of 1 nm/s . To resolve better the peaks, a lower sweep speed in combination with a better alignment would've been necessary.

For tables 4 and 5, the following notation is useful: $C = 3 \sim 6 \text{ cm}^{-1}$, $E = 15 \sim 30 \text{ cm}^{-1}$,

As it is shown on the mentioned tables, our measurements are off by a few dozens of wavenumbers. Most likely the reference between the software/monochromator was not properly

adjusted before the measurements. and hence a displacement of wavenumbers occurred in our data.

For Benzene, we were able to record only the most prominent peak, located at 991.6 cm^{-1} . The rest of the peaks are labeled in the literature (see [6]) as medium or low intensity peaks. As mentioned, our alignment was not providing a precise input scattered beam to the monochromator, so little intensity for the Raman peaks of benzene was obtained.

Table 4: Literature and measured Raman bands of carbon tetrachloride (CCl_4).

Symmetry	Mode	NIST Raman 1/cm	Uncertainty 1/cm	Measured Raman 1/cm	Deviation 1/cm
a_1	Symmetric stretch	458.7	C	490.9	32.2
e	Degenerate deformation	217.0	C	245.7	28.7
f_2	Degenerate stretch	790.4	E	798.5	8.1
f_2	Degenerate stretch	761.7	E	—	—
f_2	Degenerate deformation	313.5	C	332.3	18.8

Table 5: Literature and measured Raman bands of Benzene (C_6H_6)

Symmetry	Mode	NIST Raman 1/cm	Uncertainty 1/cm	Measured Raman 1/cm	Deviation 1/cm
a_{1g}	CH stretch	3061.9	C	—	—
a_{1g}	Ring stretch	991.6	C	1028.9	37.3
a_{2g}	CH bend	1326.0	E	—	—
e_{1g}	CH bend	848.9	C	—	—
e_{2g}	CH stretch	3046.8	C	—	—
e_{2g}	Ring stretch	1606.4	E	—	—
e_{2g}	Ring stretch	1584.6	E	—	—
e_{2g}	CH bend	1178.0	C	—	—
e_{2g}	Ring deformation	605.6	C	—	—

4.3.2 peak polarization

By observing figures 14 and 16, it is evident that the peak we measured at 490.9 cm^{-1} corresponds to strongly polarised Raman scattering. Approximating from the values recorded

$$\rho = \frac{I_{\perp}}{I_{\parallel}} \approx \frac{0.004}{0.030} = 0.13 \quad (22)$$

comparing with what was discussed in section 1.5, it is indeed strongly polarized.

The rest of the peaks recorded for CCl_4 are not strongly polarized, and from the graphs it can be estimated that $\rho \geq \frac{3}{4}$ for these.

Similarly, the only peak measured for benzene is only really visible on figure 20 which indicates strong polarization without any calculation needed.

[6] also reports the same polarization degrees as in our experiment

4.4 unknown samples

The spectra of both unknown samples was taken without the analyzer, so the depolarization ratio cannot be discussed.

To find out what molecule(s) are in the sample, the [Search for Species Data by Vibrational Energy Value](#) of the NIST was used. After introducing the wavenumbers recorded in the

experiment, a comparison between many molecules was done. given that we're lacking the polarization of the peaks, it's harder to pinpoint the actual content of the cuvette. Many molecules and peaks were looked after and a lot of the search filters were applied. Ultimately, we suspect that our spectra is heavily shifted, and we couldn't identify either of the two samples.

If we dismiss the assumption that the spectra is shifted, then the spectra for sample 1 (figure 21) contains two of the strong peaks of Borine carbonyl $\text{CH}_3^{11}\text{BO}$, albeit slightly shifted from where they should be (2380 cm^{-1} recorded at 2384 cm^{-1} and 2169 cm^{-1} recorded at 2145 cm^{-1}). However, the second value is still far enough to classify it as such.

Regarding the second sample, it was originally thought the peaks looked similar to bromoform raman peaks, but upon further inspection, the relative intensities between peaks do not match.

We conclude this discussion suggesting that narrowing down the sample would've been easier if polarization data had been recorded during the experiment, as well as a more careful alignment of the analog/software calibration of the wavenumber.

4.5 Determination of the ethanol concentration of an unknown sample

Upon further inspection, we notice that all of the ethanol samples (Figures 23 to 26) show no clear correlation between peak intensities or position. It is unclear how to proceed to make this four spectra a ruler for ethanol concentration in unknown samples. Adding the fact that most likely our unknown spectra is heavily shifted, we opted to avoid making poorly-founded calculations of ethanol concentration.

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